

BCA(Semester-II) Examination, 2022

(Session : 2020-23)

COMPUTER APPLICATION

[Paper Code : BCA-202]

(Computer Architecture)

Time : Three Hours

[Maximum Marks : 80]

Note: Candidates are required to give their answer in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. What is fixed point representation? Explain 1's and 2's complement.
2. Explain modes of Input/output data transfer or mode of data transfer.
3. Convert each of the following expressions into sum of products and product of sum:
(a) $(A+B+C) (B+C'+D)$
(b) $X'+X(X+Y')(Y+Z')$
4. Explain K-map with example.
5. Design a four-bit binary synchronous counter with D flip-flops.
6. Explain ripple counter and synchronous counter.
7. Explain isolated versus memory mapped input/output.
8. Explain auxiliary memory, magnetic disks and magnetic tape.
9. Explain cache memory, hit ratio and writing into cache.
10. Describe the following terminology:
(a) Data transfer
(b) Register
(c) Associative memory
(d) Semiconductor memories

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